THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The magnetic effect of electric current was discovered by: 1

(a) Alessandro Volta **(b)** Luigi Galvani
(c) Michael Faraday **(d)** Hans Christian Oersted

2. The strength of an electromagnet increases when: 1

(a) the number of turns increases **(b)** the wire is shortened
(c) the core is removed **(d)** current is reduced

3. A crane in a scrap yard uses an electromagnet because: 1

(a) it can be switched on and off **(b)** it attracts all materials
(c) it is cheap to melt **(d)** it never loses magnetism

4. Earth's magnetic field protects us from: 1

(a) earthquakes **(b)** gravity
(c) harmful particles from space **(d)** all rain

5. A bulb filament glows because it is: 1

(a) charged by friction **(b)** magnetised
(c) cooled by current **(d)** heated by electric current

6. Why should we not touch a nichrome wire for long when current passes? 1

(a) it becomes magnetic **(b)** it becomes hot
(c) it becomes soft **(d)** it becomes cold

7. Which scientist noticed a frog's leg twitching with two metals? 1

(a) Oersted **(b)** Faraday **(c)** Alessandro Volta **(d)** Luigi Galvani

8. The longer lead of an LED should connect to the: 1

(a) lemon only **(b)** negative terminal
(c) positive terminal **(d)** iron core

9. The electrolyte in a dry cell is a: 1

(a) metal rod **(b)** dry powder only **(c)** moist paste **(d)** gas

10. Disposing of used batteries and electronics properly is called: 1

(a) e-waste recycling **(b)** composting
(c) boiling **(d)** fermentation

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Who discovered the magnetic effect of electric current and in which year? 2

12. What is an electromagnet? 2

13. Name two factors on which the strength of an electromagnet depends. 2

14. How can the polarity of an electromagnet be reversed? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe the activity that shows a current-carrying wire deflects a compass needle. 3

16. Explain how to make a simple electromagnet. 3

17. State the factors that affect the strength of an electromagnet with the effect of each. 3

OR

Why are electromagnets used in cranes instead of permanent magnets? 3

18. List the factors on which heat produced in a wire depends. 3

OR

A student passes current through a nichrome wire and then a copper wire of same size. Predict and explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain Oersted's observation and show how an electromagnet is made. Name three factors that affect its strength. 5

OR

Describe how a lifting electromagnet in a scrap yard works and why it is better than a permanent magnet. 5

20. Explain the heating effect of current. Write factors that affect heat produced and give four applications. 5

OR

Describe an activity to show that nichrome wire gets hotter with two cells than with one cell. 5

END OF QUESTION PAPER — Total: 40 Marks

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